Factors, Multiples & Primes

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

A) Easy: List all the factors of 12.
Method & answer: 1, 2, 3, 4, 6, 12.

B) Standard: What is the Highest Common Factor (HCF) of 18 and 24?
Method & answer: Factors of 18: 1,2,3,6,9,18. Factors of 24: 1,2,3,4,6,8,12,24. Common: 1,2,3,6. HCF = 6.

C) Advanced: What is the Lowest Common Multiple (LCM) of 6 and 8?
Method & answer: Multiples of 6: 6,12,18,24... Multiples of 8: 8,16,24... First common multiple = 24. LCM = 24.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. List the factors of 8.
2. List the factors of 15.
3. List the factors of 20.
4. What are the first 5 multiples of 3?
5. What are the first 5 multiples of 7?
6. Is 17 a prime number?
7. Is 21 a prime number?
8. List the prime numbers between 1 and 10.
9. What is the smallest multiple of 9 that is greater than 40?
10. List the factors of 18.
11. Is 2 the only even prime number?
12. What are the first 4 multiples of 11?
13. List the factors of 24.
14. Which of these is a multiple of 6: 16, 18, 20?
15. Which of these is a factor of 30: 4, 5, 7?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the HCF of 36 and 48.
2. Find the HCF of 42 and 56.
3. Find the LCM of 9 and 12.
4. Find the LCM of 6 and 14.
5. Write 84 as a product of prime factors.
6. Write 90 as a product of prime factors.
7. List all the common factors of 36 and 48.
8. What is the smallest common multiple of 6 and 8?
9. Is 119 a prime number? (Hint: check 7.)
10. Which of these numbers is prime: 51, 53, 57?
11. Find the HCF of 60 and 84.
12. Find the LCM of 8 and 10.
13. List all factor pairs of 60.
14. Is 108 divisible by 9? (Use the digit-sum rule.)
15. What is the LCM of 4, 6 and 9?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two bells ring every 12 seconds and 18 seconds. If they ring together now, after how many seconds will they next ring together?

2. A baker has 48 muffins and 60 cookies. She wants identical bags with no items left over and the same number of each item per bag. What is the greatest number of bags she can make?

3. What is the smallest number that is a multiple of both 6 and 10?

4. A number less than 50 is a multiple of both 4 and 6. What is the largest such number?

5. Find a prime number between 40 and 50 that is one more than a multiple of 10.

6. Sam says "All prime numbers are odd." Give a number that proves Sam wrong.

7. What is the HCF of 45, 60 and 75?

8. Find the LCM of 8 and 12.

9. A rectangular garden is 24 m by 36 m and is to be divided into identical square plots with no space left over. What is the largest possible size of each square plot?

10. A prime number lies between 90 and 100. What is it?

11. Two trains leave a station together at 9:00am, one every 15 minutes and one every 20 minutes. When will they next leave together?

12. Write 60 as a product of prime factors.

13. Find two prime numbers that add up to 24.

14. A school wants to arrange 90 chairs and 126 chairs (separately) into rows of the same length, using all the chairs and the greatest row length common to both. What is that row length?

15. What is the smallest number greater than 1 that is both a perfect square and has exactly 3 factors?